Exam 1

September 25, 2026 | 50 minutes

Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

Name: _____

1. Domain and intercepts [5 points]

Let

$$h(x) = \frac{x-4}{x+2}.$$

(a) State the domain in interval notation. **[1 point]**

(b) Find the x-intercept. Give an ordered pair. **[2 points]**

(c) Find the y-intercept. Give an ordered pair. **[2 points]**

Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

2. Piecewise function **[5 points]**

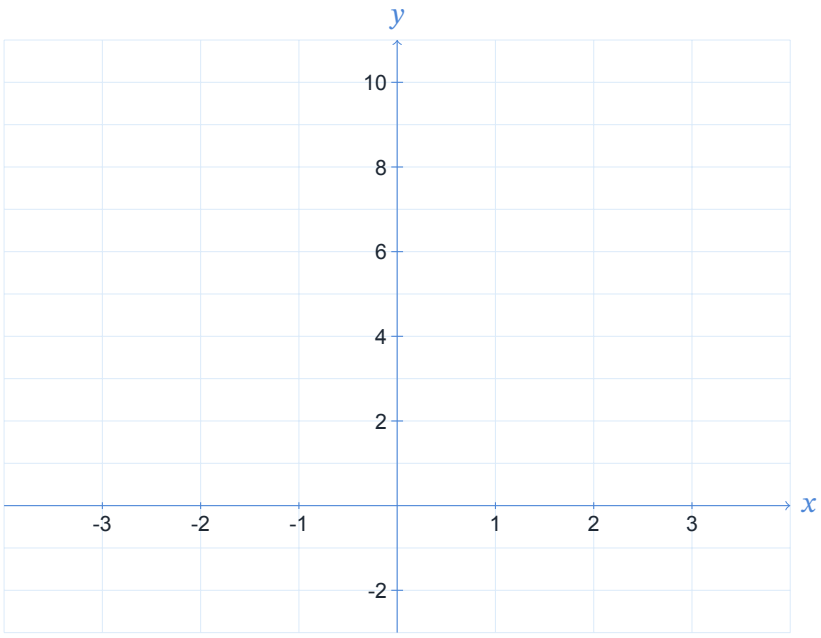
Let

$$f(x) = \begin{cases} x + 3, & x < 1, \\ x^2 - 2, & x \geq 1. \end{cases}$$

(a) Find $f(-1)$. **[1 point]**

(b) Find $f(1)$. **[1 point]**

(c) Sketch the graph for $-3 \leq x \leq 3$. Clearly mark the open and closed points at $x = 1$. **[3 points]**



Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

3. Even, odd, or neither **[5 points]**

Consider

$$g(x) = x^3 - 4x.$$

(a) Find and simplify $g(-x)$. **[2 points]**

(b) Decide whether g is even, odd, or neither. Justify using your result from (a). **[3 points]**

Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

4. A line through two points **[5 points]**

A line passes through $(-2, 5)$ and $(4, -7)$.

(a) Find its slope. **[2 points]**

(b) Write its equation in the form $y = mx + b$. **[3 points]**

Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

5. Solve by factoring **[5 points]**

Solve

$$3x^2 - 10x + 3 = 0.$$

(a) Compute ac and find two integers with product ac and sum b. **[1 point]**

(b) Split the middle term and factor by grouping. **[2 points]**

(c) Use the zero-product property to solve the equation. **[2 points]**

Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

6. Use the quadratic formula **[5 points]**

Consider

$$2x^2 + 4x - 1 = 0.$$

(a) Find the discriminant. State the number of distinct real solutions. **[2 points]**

(b) Use the quadratic formula to solve. Simplify your exact answers. **[3 points]**

Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

7. Vertex and intercept **[5 points]**

Let

$$q(x) = -x^2 + 4x + 1.$$

(a) Find the vertex and the equation of the axis of symmetry. **[2 points]**

(b) State the maximum or minimum value of q. **[1 point]**

(c) Find the y-intercept. Give an ordered pair. **[2 points]**

Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

8. Read a polynomial in factored form **[5 points]**

Let

$$p(x) = -2(x+2)^2(x-3).$$

(a) State the degree and leading coefficient. **[1 point]**

(b) List the real zeros and their multiplicities. **[2 points]**

(c) State both end behaviors using limit notation. **[2 points]**

Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

9. Sketch a polynomial **[5 points]**

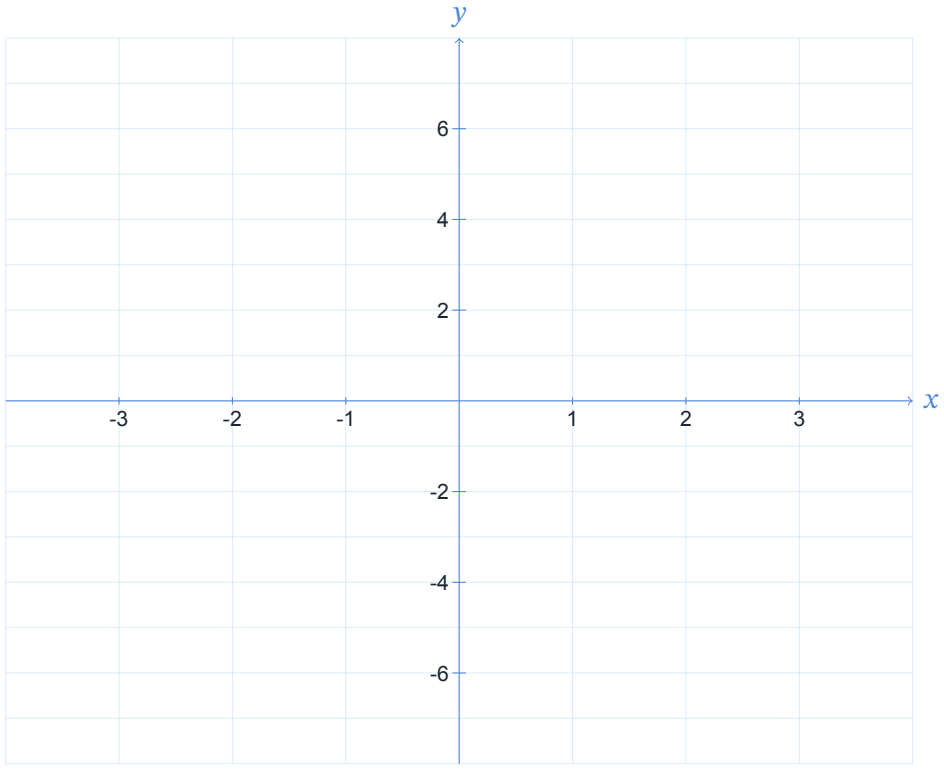
Let

$$P(x) = (x + 2)(x - 1)^2.$$

(a) Find the y-intercept. **[1 point]**

(b) At each zero, state whether the graph crosses or touches the x-axis. **[1 point]**

(c) Sketch the graph, labeling its intercepts and showing both end directions. An exact turning-point calculation is not required. **[3 points]**



Show all work. Give exact answers. **Circle your answer.** Total: 50 points.

10. Build a polynomial **[5 points]**

Find a polynomial of least degree with a zero at -1 of multiplicity 2, a zero at 3 of multiplicity 1, and $f(0) = 6$.

(a) Write its factored form using an unknown leading coefficient a. **[2 points]**

(b) Find a and write the final polynomial in factored form. **[3 points]**